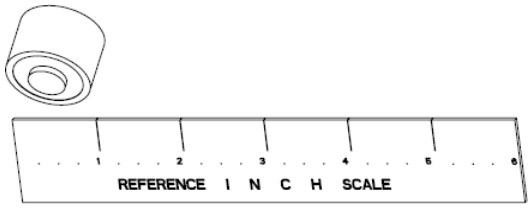
INCH-POUND
MIL-PRF-32271/9A
03 JUNE 2025
SUPERSEDING
MIL-PRF-32271/9
30 OCTOBER 2008

PERFORMANCE SPECIFICATION SHEET

BATTERY, NON-RECHARGEABLE, LITHIUM

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product herein shall consist of this specification and MIL-PRF-32271.



See NOTES for nominal dimensions; see applicable drawing for dimensional requirements.

FIGURE 1. General view, M32271/9 battery shape

For clarity, MIL-PRF-32271 tables are labeled with capital Roman numerals (e.g. IX) and MIL-PRF-32271/9A tables are labeled with lower case Roman numerals (e.g. ix).

REQUIREMENTS (see 1.2 for Type, Class, and Features descriptions):

Type	I
Class	1
Features	A
PIN	M32271/9-11A

Distribution Statement A. Approved for public release: distribution is unlimited.
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Specification requirements: The following requirements of MIL-PRF-32271, identified therein as "when specified", are applicable as indicated below:

TABLE i. *"When specified" requirements*

Requirement	Requirement Paragraph	Verification Paragraph	Applicability by PIN Suffix (M32271/9-)
			11A
Parallel discharges	*Tables II, III, IV, IX, X, XI, & XII*		*No*
Additional material requirements	*3.3.5*	*4.6.1.11*	*No*
Parallel cell arrangements	*3.4*		*No*
Connectors	*3.4.5*	*4.6.1.7*	*No*
Battery charger connection	*3.4.5.1*	*4.6.1.7.1*	*No*
Connection integrity	*3.4.5.2*		
Static	*3.4.5.2*	*4.6.1.7.2*	*No*
Dynamic	*3.4.5.2*	*4.6.1.7.3*	*No*
Terminal integrity	3.4.6		
Socket Strength	*3.4.6.1*	*4.6.1.8.1*	*No*
Terminal Strength	3.4.6.2	4.6.1.8.2	Yes
Complete discharge device	*3.4.7*	*4.6.1.9*	*No*
State of charge device	*3.4.8*	*4.6.1.10*	*No*
State of charge data output	*3.4.8e*	*4.6.1.10*	*No*
Additional design and construction requirements	*3.4.11*	*4.6.1.12*	*No*
Cell charging	*3.5.3*	*4.6.2.3*	*No*
Nail penetration	*3.5.4*	*4.6.2.4*	*No*
Parallel string charge protection	*3.5.6*	*4.6.2.6*	*No*
Cell series string short circuit	*3.5.7*	*4.6.2.7*	*No*
Battery charge protection	*3.5.12*	*4.6.2.12*	*No*
Over-current protection	*3.5.14.1*	*4.6.2.14*	*No*
Over-temperature protection	*3.5.15*	*4.6.2.15*	*No*
Projectile	3.5.16	4.6.2.17	Yes
Additional safety requirements	*3.5.18*	*4.6.2.18*	*No*
Capacity LR	*3.6*	*4.6.4.1.5*	*No*
Capacity LRT	*3.6*	*4.6.4.1.9*	*No*
Capacity IDC	*3.6*	*4.6.4.1.10*	*No*
Surface temperature	*3.6h*	*4.6.4.1*	*No*
Immersion	3.7.8	4.6.5.8	Yes
Watertight integrity	*3.7.9*		
Shallow	*3.7.9*	*4.6.5.9*	*No*
Deep	*3.7.9*	*4.6.5.10*	*No*
Electromagnetic interference	*3.7.10*	*4.6.5.11*	*No*
Electrostatic discharge	*3.7.11*	*4.6.5.12*	*No*
Additional environmental requirements	*3.7.12*	*4.6.5.13*	*No*

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TABLE i. *"When specified" requirements* – Continued

Requirement	Requirement Paragraph	Verification Paragraph	Applicability by PIN Suffix (M32271/9-)
			11A
DO NOT CHARGE! marking	*3.8.1.2b*		*No*
Complete discharge device marking/label	*3.8.4*		*No*
State of charge marking	*3.8.5*		*No*
Simplified battery label	*3.8.7*		*No*
Additional markings	*3.8.8*		*No*

Dimensions, marking and configuration

Battery - Drawing A3315886
Battery Connector – Not Applicable
Mating Connector – Not Applicable
Battery Charger Connector – Not Applicable

Battery voltages (3.4.4 & 4.6.1.5):

Battery open-circuit voltage (3.4.4.1 & 4.6.1.5.1):
Maximum – 3.3 volts
Minimum – 3.0 volts

Battery closed circuit voltage (3.4.4.2 & 4.6.1.5.2):
For capacity, initial voltage delay and battery closed circuit voltage:
Minimum – 2.0 volts

Cell closed circuit voltage test (3.4.4.3 & 4.6.1.3): Minimum - 2.0 volts per cell

Maximum weight (3.1 & 4.6.1.6):

1.1 ounces (31.2 g)

Capacity test requirements (3.6 & 4.6.4):

Test	Initial voltage delay Time (MAX)	to Volts	Minimum Capacity	Storage Period 1-Week	Storage Period 4-Week	Cut-off Volts
I	1 second	2.0	20.0 hours			2.0
L	1 second	2.0	11.0 hours			2.0
H	1 second	2.0	18.4 hours			2.0
IT	1 second	2.0		18.6 hours	18.0 hours	2.0
LT	1 second	2.0		10.2 hours	9.9 hours	2.0
HT	1 second	2.0		17.1 hours	16.6 hours	2.0

Abuse test pulse discharge minimum capacity requirement (3.5.12 & 4.6.2.12i):

0.93 ampere-hours

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METHODS OF EXAMINATION AND TEST:

Cell closed-circuit voltage test: When cells are tested as specified in 4.6.1.3, load each cell with 100 milli-amperes constant current, 20 ohms resistance, or 200 milli-watts constant power.

Battery closed-circuit voltage test: When tested as specified in 4.6.1.5.2, load each battery with 100 milli-amperes constant current, 20 ohms resistance, or 200 milli-watts constant power.

Abuse test, pre-discharge: When tested as specified in 4.6.2.13a, discharge with a load of 50 milli-amperes for 10 hours.

Abuse test, pulse discharge: When tested as specified in 4.6.2.13l, discharge with a load of 55 milli-amperes for 1 minute followed by 40 milli-amperes for 4 minutes, cycled continuously to 2.0 volts.

Immersion: When tested as specified in 4.6.5.8, apply a load of 10 milli-amperes during the storage and immersion time periods specified.

Capacity tests (4.6.4):

Test	Discharge Rate	Duty Cycle
I	50 milli-amperes	Continuous discharge to zero volts, followed by 50 milli-amperes forced discharge for 5 minutes
L	50 milli-amperes	Continuous discharge to cut-off voltage
H	50 milli-amperes	Continuous discharge to cut-off voltage
IT	50 milli-amperes	Continuous discharge to cut-off voltage
LT	50 milli-amperes	Continuous discharge to cut-off voltage
HT	50 milli-amperes	Continuous discharge to cut-off voltage

NOTES:

(This section contains information of a general nature which may be helpful, but is not mandatory)

Intended use: This specification covers non-rechargeable lithium batteries of the non-reserve type composed of electrochemical cells using lithium metal in the anode. They are intended for use in military communications and electronics equipment.

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Nominal ratings[1]: The following are the nominal ratings for the battery described by this specification sheet. They are provided for information purposes.

Battery PIN:	**M32271/9-11A (Li/MnO_2)**
Overall Dimensions:	0.749 in L x 1.0 in DIA
MAX Weight:	1.1 ounces (31.2 grams)
Voltage Range:	2.0-3.3
Nominal Capacity: (in ampere-hours)	1 Amp-hr
Nominal Energy: (in watt-hours)	2 Watt-hrs
MAX rated power output:	125 milli-watts
MAX continuous load rating:	50 milli-amperes
MAX pulse load rating:	55 milli-amperes
Instantaneous trip rating:	Not applicable
Operating temperature range:	-4 to 130ºF (-20 to 55ºC)
Storage temperature range:	-40 to 160ºF (-40 to 71ºC)
MAX abusive temperature: (non-operating)	195ºF (91ºC)

Other data:

Military Type Designations. The military type designation that relates to the PIN covered by this specification sheet is as follows: The BA-5367()/U designation has been used for the Type I PIN. It should be noted that the Type II BA-5567()/U has been replaced by the BA-5367()/U.

This battery was exempt from the transportation requirements of 49 CFR 173.185 until January 2008; now it is subject to the test requirements therein beginning in October 2009. Class 9 shipping labels typically won't be needed; consult with 49 CFR 173.185 for package quantity/size limitations. It is a single-cell battery that has a solid cathode and contains less that 1.0 grams of lithium per cell/battery. It should be noted that all non-rechargeable lithium batteries are restricted from shipment as cargo aboard passenger aircraft within, entering, or leaving the US.

This battery does not have a complete discharge device. It is considered Non-Hazardous Solid Waste per the Resource Conservation and Recovery Act (RCRA). However, it is considered Hazardous Waste in states using Bioassay rules. Check with your local disposal office for rules that apply.

The US Army CECOM Life Cycle Management Command (LCMC) publishes guidelines for the design of battery compartments for devices using both battery types I covered by this specification sheet: Technical Bulletin 43-6135. Please contact the CECOM LCMC Directorate for Safety at usarmy.APG.cecom.mbx.amsel-sf-sec-cecom-lcmc--ds@army.mil for further information.

[1] Capacity and energy delivered by a battery are significantly affected by usage conditions, such as temperature and loads applied. If you have any questions about use of this battery in a particular device or circumstance, please visit the following web site (contact info is posted): https://battery.army.mil.

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Navy safety tests of NAVSEA S9310-AQ-SAF-010 are required for this battery during first article testing. The test data provided from the testing will be used to evaluate applications for specific using devices and Navy platforms. Devices using this battery may require US Navy Safety Approval prior to use by Department of Navy users unless such approval has already been granted. Please consult with NAVSEA Instruction 9310.1 for further information.

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Custodians:
Army – CR
Navy – SH
Air Force – 99
DLA – GS

Review activities:
Navy – AS
Air Force – 71

Preparing activity:
Army – CR
(Project Number 6135-2025-010)

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